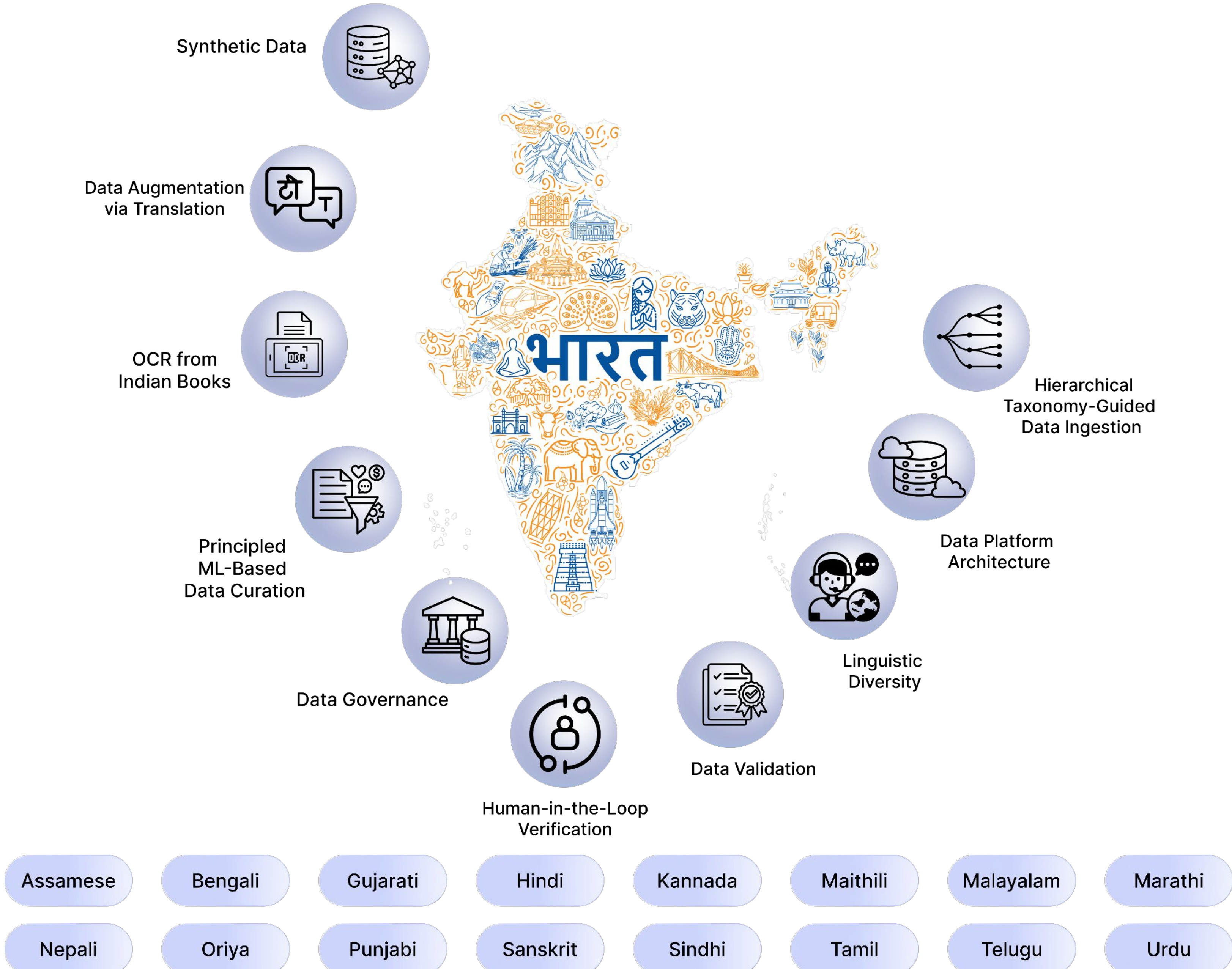


Bharat Data Sagar

Bharat Data Sagar is India's unified data refinery, powering culturally-aware AI with diverse, inclusive datasets. It bridges the digital divide to build AI that truly understands and reflects India — enabling sovereign, scalable innovation across languages and regions.

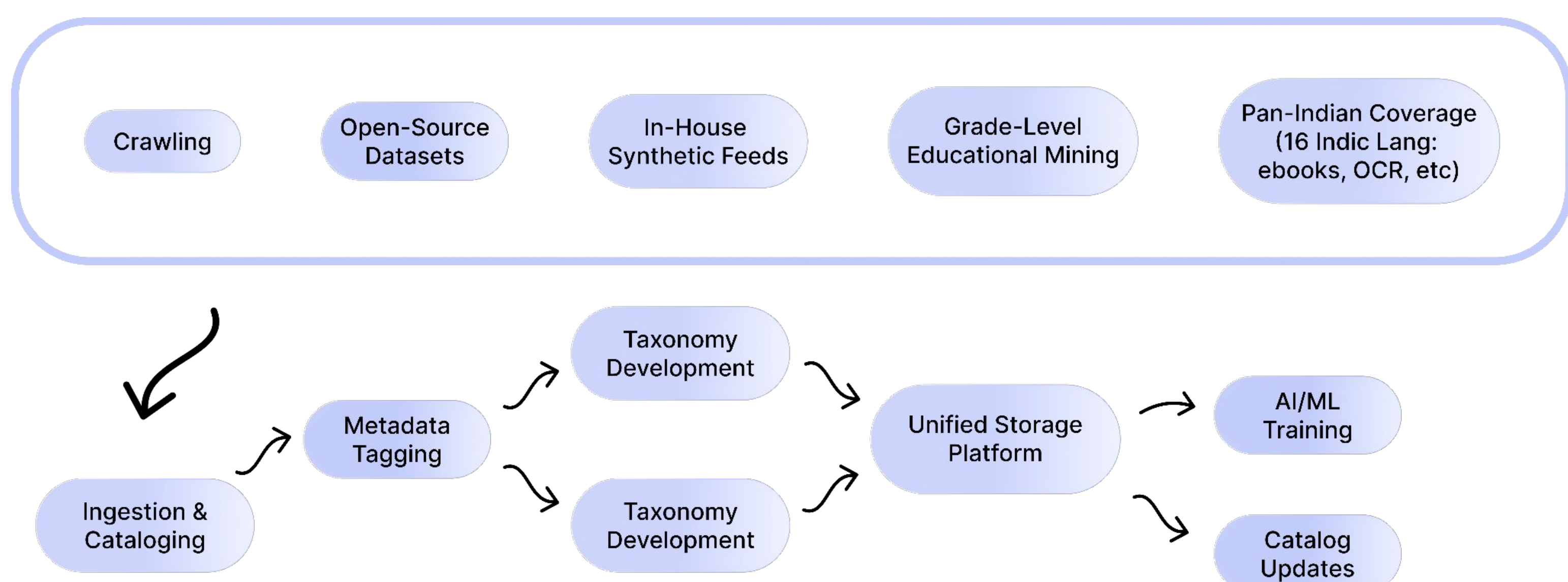


Building the Foundations of Bharat Data Sagar

At the heart of Bharat Data Sagar lie seven core data-engineering pillars—each meticulously designed to source, transform, augment, govern and validate India's rich linguistic and cultural heritage, powering AI that truly understands every region, language and domain.

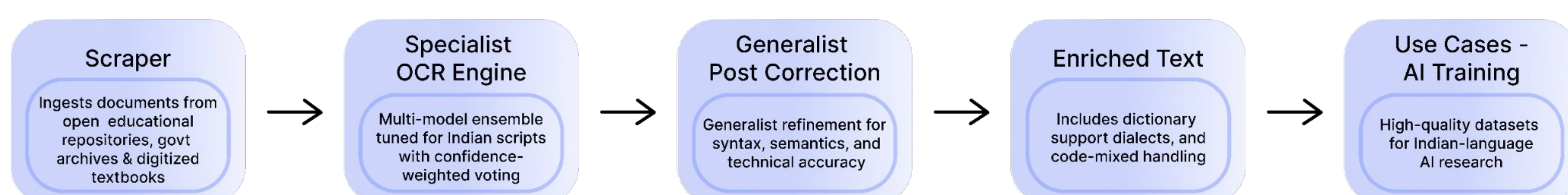
• Web Crawling & Data Sourcing

Our multi-source acquisition system harvests internet-scale content in English and 16 Indic languages—continuously enriching the data lake with curricula, vernacular media, and web trends to fuel culturally-aware AI at scale.



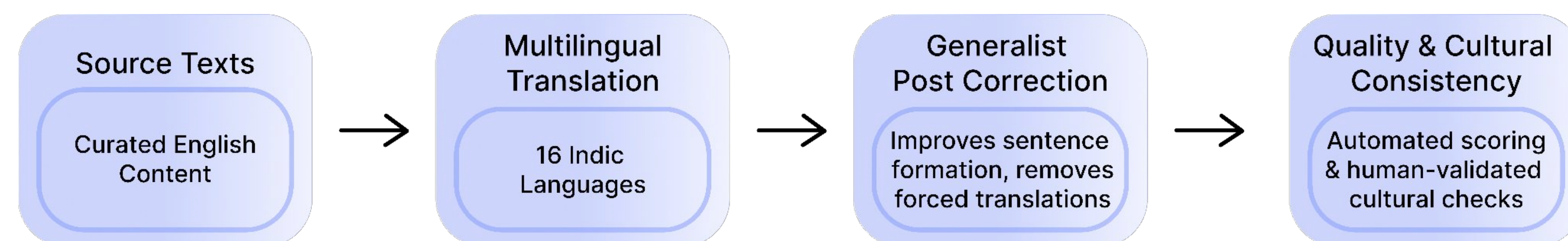
• Ensemble-Based Ocr System For Multilingual Document Digitization

Our ensemble-based OCR system combines specialist engines with LLM/VLM-powered post-correction and custom dictionaries to achieve 96%+ accuracy across 12+ Indian scripts—preserving linguistic nuance at scale for millions of pages monthly.



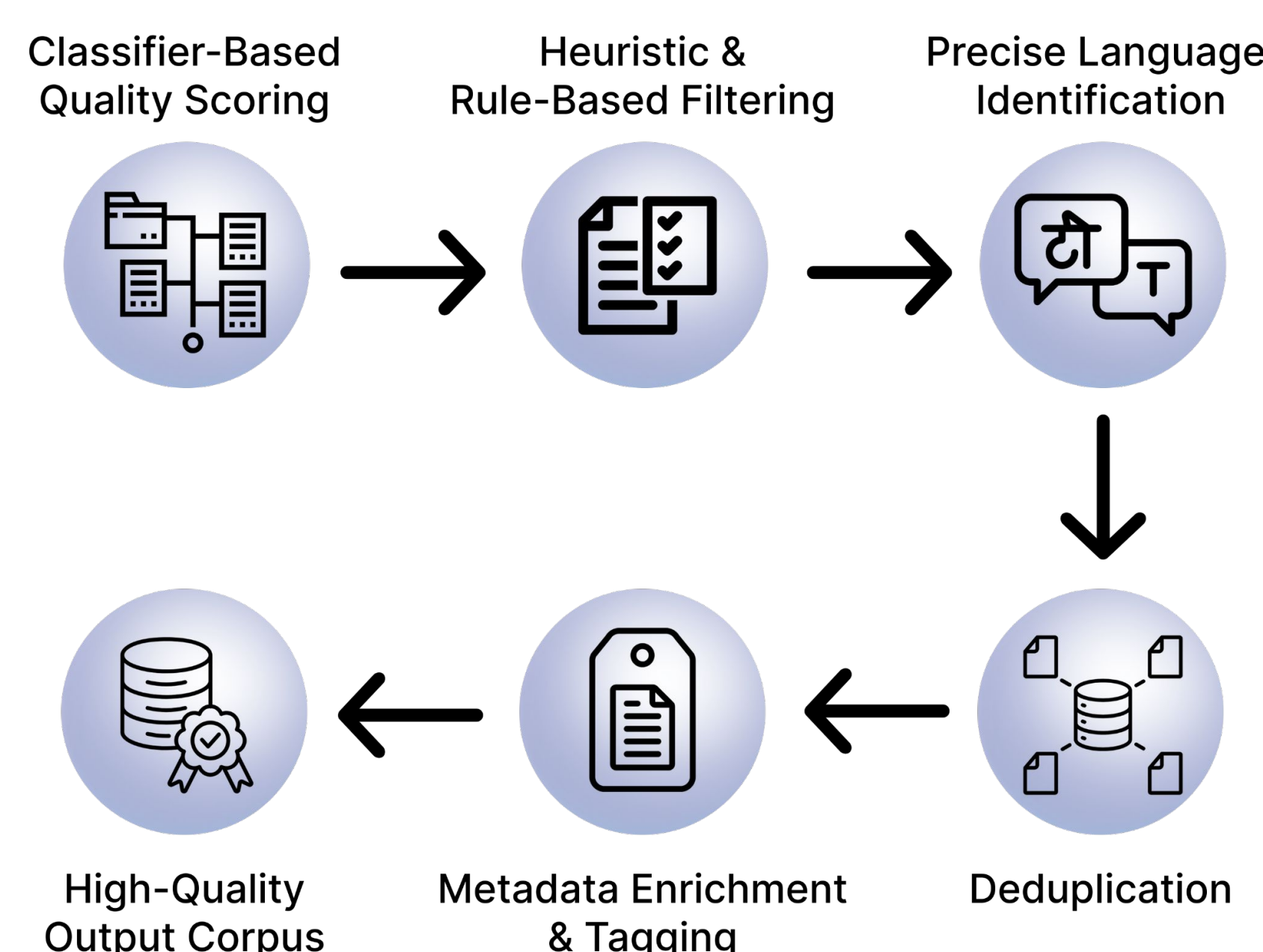
• Data Augmentation Via Translation & Synthetic Generation

Our framework expands Indian language resources by combining multilingual translation with Indian virtual persona-driven synthetic data generation across 16 Indic languages—preserving idioms and dialects through lexicon-aware workflows to build culturally rich, inclusive datasets for AI.



Data Curation & Preprocessing

Our curation pipeline cleans raw content using classifier, heuristic, and rule-based filters via NVIDIA NeMo Curator. fastText models score

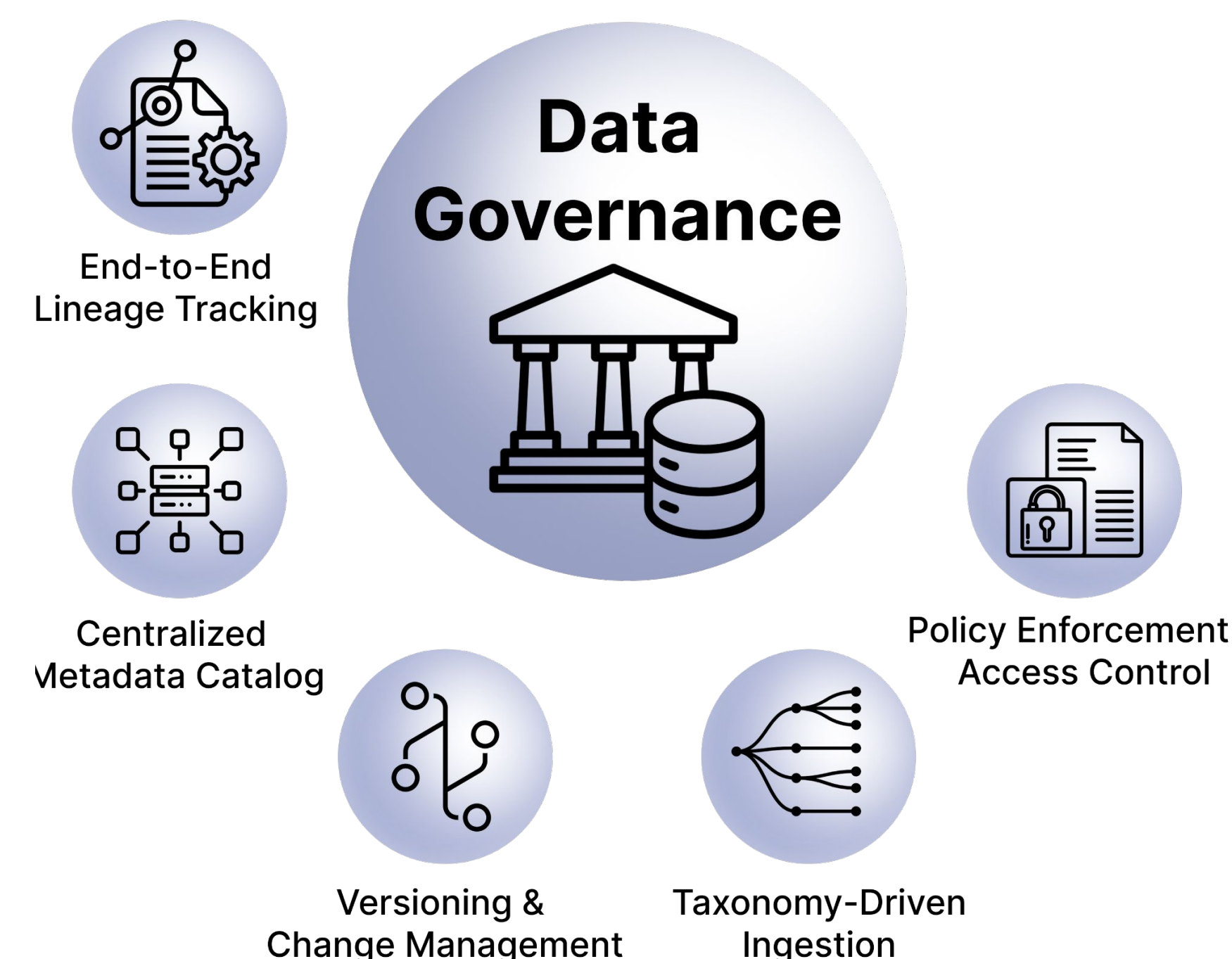


documents for quality, while language ID ensures accurate tagging of 16 Indian languages. Unicode and Indic normalization fix encoding and script issues. Deduplication and PII removal protect privacy and reduce redundancy. Each dataset is enriched with metadata like domain, language, and quality tier for easy discovery and governance.

Data Governance

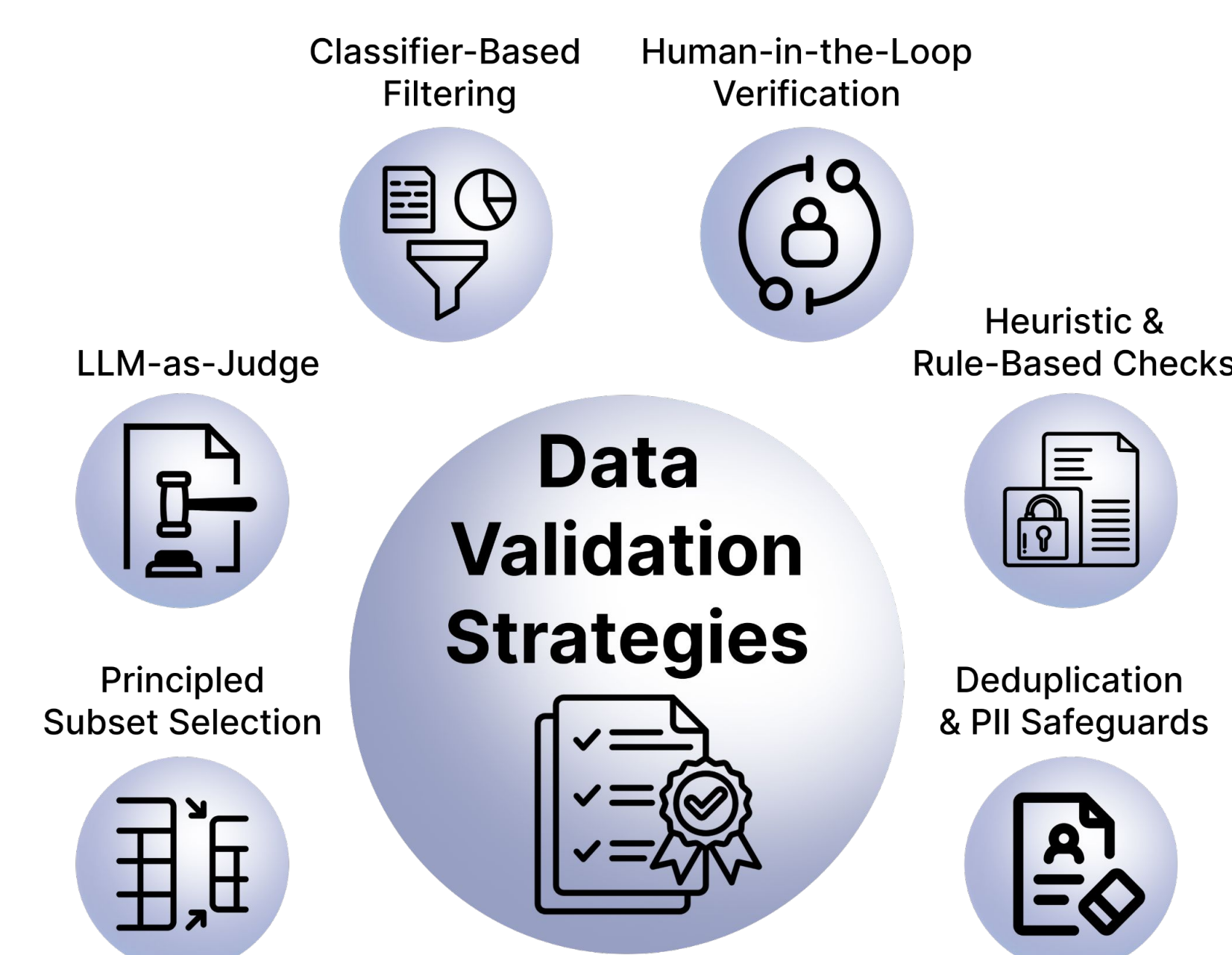
Our data governance framework ensures secure, compliant, and traceable data pipelines. Assets are tagged at source with rich metadata using hierarchical taxonomies. Every transformation is auto-logged for full lineage tracking. A centralized catalog (via DataHub/Atlas) enables self-serve discovery with clear schema context. Role-based access, integrated with LDAP/Okta, enforces security

and prevents unauthorized exports. GDPR-style consent tracking, SOC 2 controls, and versioned datasets ensure compliance, auditability, and reproducibility.



Data Validation Strategies

Our validation framework ensures dataset quality through layered checks. We use diversity-aware sampling to cover all language-domain slices, with



LLMs flag errors with confidence scores. Expert reviewers verify samples and feed corrections back into training. Classifiers and rules filter out low-quality or malformed content. Deduplication and PII redaction protect privacy and reduce bias. Dashboards and alerts track quality and trigger rechecks, ensuring every dataset meets high standards.

Data Platform Architecture

Bharat Data Sagar's architecture combines JuiceFS for scalable storage, DataHub for rich metadata management, and DuckDB for fast in-place analytics. It's fully containerized, CI/CD-driven, and monitored with Kubernetes and Prometheus—ensuring reliable, flexible, and high-performance data operations.

